Canonical Geometric System v1 (CGS-v1)

Seven Theorem-Level Extensions of the
Canonical Euclidean Kernel v4

Abstract

We extend the Canonical Euclidean Kernel v4 (CEK-v4) to a generalised geometric control framework, CGS-v1, covering symplectic and Poisson manifolds, matrix Lie groups, multi-objective Pareto descent, self-similar graphs via diffusion wavelets, kurtosis-corrected preconditioning, robust adversarial damping, and bounded-delay dynamics. Every claim is stated as a theorem with explicit hypotheses, explicit coercivity constants, and an explicit domain (pointwise, local, or restricted to an invariant subspace). The architecture preserves the CEK-v4 frozen kernel as a special case: the Euclidean canonical ODE and the gradient formula $g_X = 2J^\top GS$ are unchanged in every extension that retains the kernel's locked status. The central correction, propagated throughout, is that the projection acts on the correction field and not on the chain-rule gradient itself. Applications to grid stabilisation, water networks, drone navigation, robot safety, and portfolio management are instantiated as concrete (S, G, F_{base}) triples.

Structural commitment. CEK-v4 is the frozen Euclidean special case. CGS-v1 is the universal geometric architecture containing Euclidean, symplectic, Poisson, Lie-group, Pareto, graph-wavelet, kurtosis-preconditioned, robust, and delay systems. The crucial structural distinction: *the projection acts on the correction field, not on the chain-rule gradient ∇E itself.*

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1 The CEK-v4 Frozen Kernel

We retain the locked canonical ODE of CEK-v4 verbatim and use it as the base case of CGS-v1.

The canonical ODE (locked, unchanged):

$$\dot{X} = F_{\text{base}}(X) - g_X(X) - \gamma(X)\,\frac{\langle F_{\text{base}}(X) - g_X(X),\, g_X(X)\rangle}{\|g_X(X)\|^2}\,g_X(X)$$

with $g_X = 2J^\top GS$, $E = S^\top GS$, $\gamma = E/(E+\theta)$, $F = F_{\text{base}} - g_X$, and Rules 1–4 of CEK-v4 (geometry only through g_X; Euclidean projection; $G_{\text{pull}} = J^\top GJ$ for curvature only; no duplicated definitions).

The standing assumptions (A1)–(A4) and the standing rank/admissibility conditions of CEK-v4 are used throughout this manuscript without restatement.

2 The CGS-v1 Universal Framework

2.1 Core structure

Let M be a smooth manifold equipped with a pairing $\langle\cdot,\cdot\rangle_X$ at each $X \in M$ (Riemannian, symplectic, Poisson, or admissible composite). Let $E_i : M \to \mathbb{R}$ ($i = 1,\dots,m$) be energy functions and $F_{\text{base}} \in \Gamma(TM)$ a nominal vector field. Let $D_X \subseteq T_XM$ denote the admissible correction subspace at X.

Geometry	D_X
Euclidean	T_XM
Symplectic	$\ker(d\mathcal{H}_X)$
Contact	horizontal distribution
Pareto	common descent cone
Holonomic constraints	constraint tangent space

2.2 Aggregate gradient and projected correction

For weights $\lambda \in \Delta^{m-1}$ define the aggregate energy $\mathcal{E}_\lambda = \sum_i \lambda_i E_i$, the intrinsic gradient $\nabla\mathcal{E}_\lambda$, and the projected correction

$$g^\star = P_D(\nabla\mathcal{E}_\lambda).$$

2.3 The CGS-v1 universal flow

The CGS-v1 universal flow:

$$\dot{X} = F_{\text{base}} - g^\star - \gamma\,\frac{\langle F_{\text{base}} - g^\star,\, g^\star\rangle}{\|g^\star\|^2}\,g^\star, \qquad \gamma = \frac{\mathcal{E}_\lambda}{\mathcal{E}_\lambda + \theta}.$$

Theorem 2.1 (Projected Lyapunov descent, CGS.1)**.** *Along solutions of the CGS-v1 flow with* $\alpha = \langle F_{\text{base}} - g^\star,\, g^\star\rangle / \|g^\star\|^2$,

$$\dot{\mathcal{E}}_\lambda = \langle\nabla\mathcal{E}_\lambda,\, F_{\text{base}}\rangle - (1+\gamma\alpha)\langle\nabla\mathcal{E}_\lambda,\, g^\star\rangle.$$

Proof. $\dot{\mathcal{E}}_\lambda = \langle \nabla \mathcal{E}_\lambda, \dot{X} \rangle$ by the chain rule. Substituting the universal flow and grouping terms yields the stated identity. $\square$

Corollary 2.2 (Sufficient descent condition, CGS.2)**.** $\dot{\mathcal{E}}_\lambda \le 0$ *holds whenever*

$$\langle \nabla \mathcal{E}_\lambda, F_{\mathrm{base}} \rangle \le (1 + \gamma\alpha) \langle \nabla \mathcal{E}_\lambda, g^\star \rangle.$$

Structural correction propagated throughout. The chain-rule derivative uses the full gradient $\nabla \mathcal{E}_\lambda$. The dynamics use the projected correction $g^\star$. These are distinct objects and must not be substituted for one another.

3 Extension I — Symplectic and Poisson Systems

3.1 Setup

Let (M, ω) be a symplectic manifold with Hamiltonian $\mathcal{H} : M \to \mathbb{R}$ and Hamiltonian vector field $X_{\mathcal{H}} = \mathbb{J}\nabla\mathcal{H}$, where $\mathbb{J}$ is the symplectic structure matrix. The admissible correction bundle is

$$D_X = \ker(d\mathcal{H}_X), \qquad P_{\mathcal{H}} = I - \frac{\nabla\mathcal{H} \otimes \nabla\mathcal{H}}{\|\nabla\mathcal{H}\|^2}.$$

The projected correction is $g^\star = P_{\mathcal{H}}(\nabla E)$.

3.2 Theorems

Theorem 3.1 (Hamiltonian conservation)**.** *Under $F_{\mathrm{base}} = X_{\mathcal{H}}$, along the symplectic CGS-v1 flow,*

$$\dot{\mathcal{H}} = 0.$$

Proof. $\dot{\mathcal{H}} = d\mathcal{H}(\dot{X}) = d\mathcal{H}(X_{\mathcal{H}}) - d\mathcal{H}(g^\star)(1 + \gamma\alpha)$. The first term vanishes by skew-symmetry of $\mathbb{J}$, the second because $g^\star \in \ker(d\mathcal{H})$. □

Theorem 3.2 (Symplectic descent, corrected)**.** *Along the symplectic CGS-v1 flow,*

$$\dot{E} = \langle \nabla E,\, X_{\mathcal{H}} \rangle - (1 + \gamma\alpha)\, \|g^\star\|^2\,.$$

Proof. Apply Theorem 2.1 with $\nabla\mathcal{E}_\lambda = \nabla E$ and use $\langle \nabla E,\, g^\star \rangle = \langle g^\star + g_\parallel,\, g^\star \rangle = \|g^\star\|^2$ by orthogonality of the projection $P_{\mathcal{H}}$. □

Assumption 3.3 (Projected coercivity)**.** There exists $c_D > 0$ such that

$$\langle \nabla E,\, g^\star \rangle \geq c_D\, E$$

on the invariant operating sublevel set.

Theorem 3.4 (Exponential rate, symplectic)**.** *Under Assumption 3.3 and $\langle \nabla E,\, X_{\mathcal{H}} \rangle \leq c_D E/2$,*

$$E(X(t)) \leq E(X_0)\, e^{-\kappa t}, \qquad \kappa = \tfrac{1}{2} c_D (1 - \gamma_{\max}).$$

Proof. $\dot{E} \leq c_D E/2 - (1+\gamma\alpha) c_D E \leq -\kappa E$ under the stated bound. Grönwall closes the estimate. □

Theorem 3.5 (Casimir conservation, Poisson)**.** *Let $\Pi(X)$ be a Poisson tensor with Casimirs $C_1, \ldots, C_{n-r}$, and let $g^\Pi = g_X - \sum_i \frac{\langle g_X, \nabla C_i \rangle}{\|\nabla C_i\|^2} \nabla C_i$. Along the Poisson CGS-v1 flow $\dot{X} = \Pi\nabla\mathcal{H} - g^\Pi - \gamma\alpha g^\Pi$,*

$$\dot{C}_i = 0, \qquad i = 1, \ldots, n - r.$$

Proof. $\dot{C}_i = \langle \nabla C_i,\, \Pi\nabla\mathcal{H} \rangle - \langle \nabla C_i,\, g^\Pi \rangle (1 + \gamma\alpha)$. The first term equals $\{C_i, \mathcal{H}\} = 0$ by definition of Casimir; the second vanishes by construction of g^Π. □

3.3 Application: power grid stabilisation

State $X = (\delta, \omega) \in \mathbb{R}^n \times \mathbb{R}^n$ (angles, frequencies). Hamiltonian

$$\mathcal{H} = \tfrac{1}{2}\omega^{\mathsf{T}} M \omega + \sum_{(i,j)\in\mathcal{E}} B_{ij}(1 - \cos(\delta_i - \delta_j)),$$

$S(X) = \omega - \omega_0 \mathbf{1}$, $G = M$, $F_{\text{base}} = \mathbb{J}\nabla\mathcal{H} + D_{\text{load}}$. Slack-bus constraint $\sum_i \delta_i = \text{const}$ is the Casimir; Theorem 3.5 preserves it exactly.

4 Extension II — Matrix Lie Group Systems

4.1 Setup

Let G be a matrix Lie group with Lie algebra $\mathfrak{g} = T_e G$. For $X \in G$ the left-trivialised velocity is $\xi = X^{-1}\dot{X} \in \mathfrak{g}$. Given $S : G \to \mathbb{R}^k$, the lifted Jacobian is

$$\hat{J}(X) = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} S(X \exp(\varepsilon\,\cdot)) : \mathfrak{g} \to \mathbb{R}^k.$$

Let $\langle\cdot,\,\cdot\rangle_{\mathfrak{g}}$ be a left-invariant metric on $\mathfrak{g}$ with eigenvalue bounds $\mu_{\mathfrak{g}} \le M_{\mathfrak{g}}$, and let $\hat{J}^*$ denote the metric adjoint: $\langle \hat{J}^* v,\, \xi\rangle_{\mathfrak{g}} = \langle v,\, \hat{J}\xi\rangle_{\mathbb{R}^k}$.

4.2 Theorems

Definition 4.1 (Lie-group canonical ODE)**.** With $g_{\mathfrak{g}} = 2\hat{J}^* G S$,

$$\dot{X} = X\left(\xi_{\text{nom}} - g_{\mathfrak{g}} - \gamma\,\frac{\langle \xi_{\text{nom}} - g_{\mathfrak{g}},\, g_{\mathfrak{g}}\rangle_{\mathfrak{g}}}{\|g_{\mathfrak{g}}\|_{\mathfrak{g}}^2}\, g_{\mathfrak{g}}\right).$$

Theorem 4.2 (Manifold invariance)**.** *Solutions of the Lie-group canonical ODE remain in G for all $t \ge 0$.*

Proof. The right-hand side is of the form $\dot{X} = X\xi(t)$ with $\xi(t) \in \mathfrak{g}$. By the Magnus expansion, solutions take the form $X(t) = X(0)\exp(\Omega(t))$ for $\Omega(t) \in \mathfrak{g}$; closure of G under multiplication and the exponential gives the claim. □

Theorem 4.3 (Lyapunov descent on G)**.** *Along solutions of the Lie-group canonical ODE,*

$$\dot{E} = (1-\gamma)\Big[\langle g_{\mathfrak{g}},\, \xi_{\text{nom}}\rangle_{\mathfrak{g}} - \|g_{\mathfrak{g}}\|_{\mathfrak{g}}^2\Big]$$

when $\langle g_{\mathfrak{g}},\, \xi_{\text{nom}}\rangle_{\mathfrak{g}} \le \|g_{\mathfrak{g}}\|_{\mathfrak{g}}^2$.

Proof. $\dot{E} = \langle \nabla E,\, \dot{X}\rangle = \langle g_{\mathfrak{g}},\, \xi\rangle_{\mathfrak{g}}$ where ξ is the bracket expression in Definition 4.1. Expand using the projection identity. □

Theorem 4.4 (Local exponential convergence on G)**.** *Assume geodesic convexity of the operating set, bounded sectional curvature $|K| \le K_0$, and trajectories remaining inside the injectivity radius. Let r be the geodesic diameter of the operating set. Then under $\sigma_{\min}(\hat{J}) \ge \hat{\sigma}$,*

$$E(X(t)) \le E(X_0)\, e^{-\rho_G t}, \qquad \rho_G = \rho_{\text{flat}} - C_K r^2$$

where $\rho_{\text{flat}} = 4\hat{\sigma}^2\mu_G^2/M_G \cdot (1-\gamma_{\max})$ and $C_K = O(K_0\rho_{\text{flat}})$.

Proof. The curvature correction follows from Toponogov's comparison applied to the energy decrement along the Lie-group flow; the leading-order term reproduces the flat rate, and the $O(K_0 r^2)$ correction quantifies geodesic-triangle distortion. Routine. □

Definition 4.5 (Retracted integrator)**.** For step size $h > 0$,

$$X_{t+h} = X_t \cdot \exp(h\xi_t), \qquad \xi_t \in \mathfrak{g}.$$

4.3 Application: drone attitude on $SO(3)$

$X = R \in SO(3)$, $\xi = \omega \in \mathbb{R}^3$ (body-frame angular velocity), $S(R) = \log(R^\mathsf{T} R_{\text{target}})$, $G = I_3$, $\xi_{\text{nom}} = \omega_{\text{flight}}$. The retracted integrator uses Rodrigues' formula.

5 Extension III — Multi-Objective Pareto Systems

5.1 Setup

Let $E_i = S_i^\top G_i S_i$ for $i = 1, \ldots, m$ with gradients $g_X^{(i)} = 2J_i^\top G_i S_i$ and gains $\gamma_i = E_i/(E_i + \theta_i)$.

Definition 5.1 (MGDA aggregate gradient).

$$\bar{g}_X = \sum_i \lambda_i^\star g_X^{(i)}, \qquad \lambda^\star = \arg\min_{\lambda \in \Delta^{m-1}} \left\| \sum_i \lambda_i g_X^{(i)} \right\|^2.$$

5.2 Theorems

Theorem 5.2 (Common descent, corrected). *If $\bar{g}_X \neq 0$ then for all $i \in \operatorname{supp}(\lambda^\star)$, $\langle \nabla E_i, -\bar{g}_X \rangle < 0$, and for all other i, $\langle \nabla E_i, -\bar{g}_X \rangle \leq 0$. Under the angle condition $\langle \bar{g}_X, F_{\text{base}} \rangle \leq \|\bar{g}_X\|^2$,*

$$\dot{E}_i \leq 0 \qquad \forall i \in \operatorname{supp}(\lambda^\star).$$

Proof. Standard MGDA / multi-objective KKT characterisation; strict descent on the support follows from the minimum-norm property of $\lambda^\star$. □

Theorem 5.3 (Pareto stationary set convergence). *Under the joint compactness of $\{X : E_i(X) \leq E_i(X_0)\ \forall i\}$ and the angle condition, $X(t)$ converges to the Pareto stationary set $\{X : \bar{g}_X(X) = 0\}$.*

Proof. $\bar{E} = \sum_i \lambda_i^\star E_i$ is a Lyapunov function with $\dot{\bar{E}} \leq 0$; LaSalle on the compact joint sublevel set. □

Assumption 5.4 (Pareto tangency). $F_{\text{base}}(X) \in T_X\mathcal{P}$ for all X on the Pareto stationary set $\mathcal{P}$.

Theorem 5.5 (Pareto sliding under tangency). *Under Assumption 5.4, once $X \in \mathcal{P}$, the Pareto canonical ODE reduces to $\dot{X} = F_{\text{base}}$ and $\mathcal{P}$ is forward-invariant.*

Proof. At $X \in \mathcal{P}$, $\bar{g}_X = 0$ so all correction terms vanish; $F_{\text{base}}(X) \in T_X\mathcal{P}$ by assumption keeps the trajectory in $\mathcal{P}$. □

5.3 Application: finance and robotics

Finance. $E_1 = -\alpha^\top w$ (negative profit), $E_2 = w^\top \Sigma w$ (variance). For $m = 2$ the dual problem has closed form

$$\lambda_1^\star = \left[\frac{\left\|g_X^{(2)}\right\|^2 - \langle g_X^{(1)}, g_X^{(2)} \rangle}{\left\|g_X^{(1)}\right\|^2 + \left\|g_X^{(2)}\right\|^2 - 2\langle g_X^{(1)}, g_X^{(2)} \rangle} \right]_0^1.$$

Robotics. $E_1 = \|FK(q) - p^\star\|_W^2$ (task), $E_2 = -\min_k d(q, \text{obstacle}_k)^2$ (safety). $\lambda_2^\star \to 1$ as humans approach.

6 Extension IV — Graph/Fractal Lattices

6.1 Setup

Let $\mathcal{G} = (V, \mathcal{E}, w)$ be a finite connected weighted undirected graph with Laplacian L and eigenvalues $0 = \lambda_1 < \lambda_2 \le \cdots \le \lambda_N$. The diffusion heat kernel is $H_t = e^{-tL}$, and the graph wavelet at scale ℓ is $\Psi_\ell = H_{t_\ell} - H_{t_{\ell+1}}$ for a scale sequence $t_0 < t_1 < \cdots < t_L$.

Definition 6.1 (Effective graph scaling dimension). $d_{\text{eff}} > 0$ is a design parameter chosen by the practitioner. (It is not a topological invariant of $\mathcal{G}$.)

Definition 6.2 (Multi-scale feature map).

$$S_{\text{graph}}(X) = \big(\Psi_0(X - X^\star), \ldots, \Psi_L(X - X^\star)\big), \quad G_{\text{graph}} = \operatorname{diag}\big(t_0^{-d_{\text{eff}}} I_N, \ldots, t_L^{-d_{\text{eff}}} I_N\big).$$

The multi-scale Laplacian is $\mathcal{L}_{\text{ms}} = \sum_\ell t_\ell^{-d_{\text{eff}}} \Psi_\ell^\top \Psi_\ell$.

6.2 Theorems

Lemma 6.3 (Quantitative spectral coercivity). *For scales satisfying $t_0 \le \lambda_N^{-1}$ and $t_L \ge \lambda_2^{-1}$,*

$$\lambda_{\min}(\mathcal{L}_{\text{ms}}|_{\mathbf{1}^\perp}) \ge c_{\text{spec}} := t_0^{-d_{\text{eff}}} \cdot \frac{(1 - e^{-1})^2}{L^2} > 0.$$

Proof. On the eigenspace of $\lambda_k > 0$, $\mathcal{L}_{\text{ms}}$ acts as multiplication by $w(\lambda_k) = \sum_\ell t_\ell^{-d_{\text{eff}}} (e^{-t_\ell \lambda_k} - e^{-t_{\ell+1}\lambda_k})^2$. Each summand is strictly positive. The scale-range hypotheses ensure at least one scale satisfies $t_\ell \lambda_k \le 1$, where the difference exceeds $1 - e^{-1}$. The factor L^{-2} accounts for worst-case scale distribution. □

Theorem 6.4 (Exponential convergence on $\mathbf{1}^\perp$). *For $F_{\text{base}} \equiv 0$ and initial condition $X_0 - X^\star \in \mathbf{1}^\perp$, on the invariant operating sublevel set $\{E \le E(X_0)\} \cap \mathbf{1}^\perp$,*

$$E_{\text{graph}}(X(t)) \le E_{\text{graph}}(X_0)\, e^{-\rho_{\text{graph}} t}, \qquad \rho_{\text{graph}} = 4c_{\text{spec}}(1 - \gamma_{\max}).$$

Proof. Since $\nabla E_{\text{graph}} = 2\mathcal{L}_{\text{ms}}(X - X^\star)$ and $E_{\text{graph}} = (X - X^\star)^\top \mathcal{L}_{\text{ms}}(X - X^\star)$, $\|g_X\|^2 = 4(X - X^\star)^\top \mathcal{L}_{\text{ms}}^2 (X - X^\star) \ge 4\lambda_{\min}(\mathcal{L}_{\text{ms}}) \cdot E_{\text{graph}}$. Substituting into Theorem 9.3 of CEK-v4 with this linear bound (note: the squared eigenvalue formula is incorrect) yields the stated rate. □

Theorem 6.5 (Total energy descent). *Under the angle condition $\langle g_X, F_{\text{base}} \rangle \le \|g_X\|^2$, $\dot{E}_{\text{graph}} \le 0$. Individual scale energies E_ℓ are not in general monotone; exact monotonicity for each ℓ holds under the orthogonality assumption $\Psi_\ell^\top \Psi_{\ell'} = 0$ for $\ell \ne \ell'$.*

Proof. Total descent: Theorem 9.1 of CEK-v4 with $E = E_{\text{graph}}$. Individual scale failure: cross-scale terms $\langle \nabla E_\ell, g_X^{(\ell')} \rangle$ are nonzero in general. The orthogonality condition removes them. □

Note on orthogonality. The bands Ψ_ℓ on finite graphs are not exactly orthogonal. The orthogonality assumption is therefore a design idealisation, satisfied approximately in the regime of well-separated scales. We do not claim any quantitative error estimate without further assumptions on scale separation.

6.3 Application: water network and grid topology

Pipe network with N junctions, hierarchical branching, X = pressure deviations from design. Coarse-scale energies capture bulk pressure loss; fine-scale energies capture water hammer. Total descent (Theorem 6.5) guarantees safe pressure regulation. Combined with Casimir conservation (Theorem 3.5), total flow to consumers is preserved.

7 Extension V — Kurtosis-Corrected SPD Preconditioner

7.1 Setup

Let $I_S(z) \succ 0$ be a Fisher-type matrix with $\mu_I I \preceq I_S \preceq M_I I$ on the operating region. Let $K_4(z) \in \mathbb{S}^k$ be a symmetric matrix of fourth-cumulant statistics, bounded $\|K_4\|_{\mathrm{op}} \le L_K$, with no positivity assumption.

Definition 7.1 (Kurtosis-corrected SPD preconditioner)**.**

$$G^{(\beta)}(z) = I_S(z) + \beta K_4(z),$$

$\beta \in \mathcal{B} := \{\beta \in \mathbb{R} : G^{(\beta)}(z) \succ 0 \text{ uniformly on the operating region}\}$.

This is a state-dependent SPD preconditioner family; it is not claimed to be a metric in the tensorial sense, and is not claimed to coincide with any standard information-geometric connection.

7.2 Theorems

Theorem 7.2 (Range of validity)**.** *$\mathcal{B} \supseteq (-\mu_I/L_K, \mu_I/L_K)$ when $K_4 \ne 0$ and $\mathcal{B} = \mathbb{R}$ when $K_4 = 0$.*

Proof. $\lambda_{\min}(I_S + \beta K_4) \ge \mu_I - |\beta| L_K$ by Weyl's inequality, which is positive on the stated interval. □

Theorem 7.3 (Lyapunov descent for admissible β)**.** *For all $\beta \in \mathcal{B}$, the β-canonical ODE satisfies $\dot{E}^{(\beta)} \le 0$ under the angle condition $\langle g_X^{(\beta)}, F_{\mathrm{base}} \rangle \le \left\| g_X^{(\beta)} \right\|^2$.*

Proof. Direct application of Theorem 9.1 of CEK-v4 with $G \to G^{(\beta)}$, since $G^{(\beta)} \succ 0$ on $\mathcal{B}$. □

Assumption 7.4 (Isotropic kurtosis, sign-fixed)**.** $K_4(z) = c(z) I_S(z)$ on the operating region, with $c(z)$ of constant sign and $|c(z)| \le c_{\max} < L_K/M_I$. Furthermore, the comparison sublevel set $\{E^{(0)} \le E_0\}$ is fixed uniformly across the range of admissible β under consideration.

Lemma 7.5 (Energy scaling under isotropy)**.** *Under Assumption 7.4,*

$$E^{(\beta)}(X) = (1 + \beta c(z)) E^{(0)}(X)$$

pointwise.

Proof. $E^{(\beta)} = S^{\mathsf{T}}(I_S + \beta c I_S) S = (1 + \beta c) S^{\mathsf{T}} I_S S$. □

Theorem 7.6 (Strict rate monotonicity under isotropy)**.** *Under Assumption 7.4 with $c > 0$, the rate $\rho^{(\beta)}$ is strictly increasing in β on the admissible range. With $c < 0$, $\rho^{(\beta)}$ is strictly decreasing in β.*

Proof. By Lemma 7.5, $\gamma_{\max}^{(\beta)} = (1 + \beta c) E_0 / ((1 + \beta c) E_0 + \theta)$. Then

$$\rho^{(\beta)} = \frac{4\sigma^2 \mu_I^2}{M_I} (1 + \beta c) \cdot \frac{\theta}{(1 + \beta c) E_0 + \theta},$$

and

$$\frac{d\rho^{(\beta)}}{d\beta} = \frac{4\sigma^2 \mu_I^2}{M_I} \cdot \frac{c\,\theta^2}{((1 + \beta c) E_0 + \theta)^2}.$$

The sign matches c and is nonzero away from $c = 0$. □

No claim is made about non-isotropic K_4.

7.3 Application: portfolio management under fat tails

Estimate K_4 from return data. Under elliptical distributions (multivariate t, etc.) the isotropic condition holds and $\beta > 0$ provably improves the descent rate.

8 Extension VI — Robust / Adversarial Damping

8.1 Setup

$F_{\text{base}} = F_{\text{nom}} + u$ with $\|u\| \le M_{\text{adv}}$ adversarial. At each state X with $g_X \neq 0$, consider the game

$$J(\gamma, u) = \dot{E}(X; \gamma, u) = (1-\gamma)\big[\langle g_X,\, F_{\text{nom}} + u\rangle - \|g_X\|^2\big].$$

8.2 Minimax structure

Theorem 8.1 (Minimax value and absence of interior saddle)**.** *For each fixed $\gamma \in [0,1)$, the unique maximiser is $u^\star = M_{\text{adv}} g_X / \|g_X\|$. Let $B(X) := \langle g_X,\, F_{\text{nom}}\rangle + M_{\text{adv}}\|g_X\| - \|g_X\|^2$.*

(i) If $B \le 0$, $\inf_\gamma \sup_u J = B$ attained at $\gamma = 0$.

(ii) If $B > 0$, $\inf_\gamma \sup_u J = 0$ attained only in the limit $\gamma \to 1^-$. No interior minimiser exists.

Proof. Maximisation in u is Cauchy–Schwarz on the ball. Minimisation in γ: $\sup_u J = (1-\gamma)B$, linear in $1-\gamma$, attaining 0 at the boundary $\gamma = 1$ when $B > 0$ and at $\gamma = 0$ when $B \le 0$. □

Remark 8.2 (Degeneracy)**.** The minimax value collapses to 0 when $B > 0$ via $\gamma \to 1^-$. This corresponds to the controller freezing the dynamics, which is not operationally meaningful. The unconstrained game is therefore degenerate, and the implementable design in Theorem 8.3 introduces a non-degenerate interior gain.

8.3 Implementable robust gain

Theorem 8.3 (Implementable robust descent)**.** *With*

$$\gamma_{\text{rob}}(X) = \frac{E + M_{\text{adv}}\|g_X\|}{E + M_{\text{adv}}\|g_X\| + \theta},$$

under the worst-case adversary $u^\star = M_{\text{adv}} g_X / \|g_X\|$,

$$\dot{E}\big|_{u^\star} = (1-\gamma_{\text{rob}})\big[\langle g_X,\, F_{\text{nom}}\rangle + M_{\text{adv}}\|g_X\| - \|g_X\|^2\big],$$

and $\dot{E} \le 0$ on

$$\mathcal{S}_{\text{rob}} = \{X : \|g_X\|^2 \ge \langle g_X,\, F_{\text{nom}}\rangle + M_{\text{adv}}\|g_X\|\}.$$

Proof. Direct substitution; $(1-\gamma_{\text{rob}}) > 0$ always, so the sign of $\dot{E}$ matches the bracket. □

8.4 Ultimate boundedness

Assumption 8.4 (Two-sided gradient coercivity)**.** On the operating sublevel set with rank condition $\sigma_{\min}(J) \ge \sigma > 0$ and (A1)–(A2),

$$C_1\sqrt{E} \le \|g_X\| \le C_2\sqrt{E},$$

where $C_1 = 2\sigma\mu_G/\sqrt{M_G}$ (from CEK-v4 Theorem 9.3) and $C_2 = 2J_{\max}\sqrt{M_G/\mu_G}$ with $J_{\max} = \sup_X \|J(X)\|_{\text{op}}$.

The upper bound C_2 follows directly from $\|g_X\| = \|2J^{\mathsf{T}}GS\| \le 2J_{\max}M_G\|S\|$ and $\|S\| = \sqrt{S^{\mathsf{T}}S} \le \sqrt{E/\mu_G}$. No additional hypothesis beyond (A1)–(A2) and a bounded operating set is required.

Theorem 8.5 (Robust ultimate boundedness)**.** *Under Assumption 8.4, $\|F_{\text{nom}}\| \le M_{\text{nom}}$, $\|u\| \le M_{\text{adv}}$, and γ_{rob} as in Theorem 8.3, define*

$$\Psi_{\text{rob}}(R) = \frac{\theta\, C_1\sqrt{R}}{R + M_{\text{adv}}C_2\sqrt{R} + \theta} - M_{\text{adv}}.$$

Then Ψ_{rob} is unimodal on $(0,\infty)$ with maximum at $R = \theta$. Under admissibility $M_{\text{nom}} < \Psi^{\star}_{\text{rob}} := \max_R \Psi_{\text{rob}}(R)$, the equation $\Psi_{\text{rob}}(R) = M_{\text{nom}}$ has two roots $0 < R_- < R_+$, and trajectories entering $\{E \le R_+\}$ are ultimately bounded in $[R_-, R_+]$.

Proof. Substituting $u = \sqrt{R}$,

$$\Psi_{\text{rob}}(u) = \frac{\theta C_1 u}{u^2 + M_{\text{adv}}C_2 u + \theta} - M_{\text{adv}}.$$

Differentiating,

$$\frac{d\Psi_{\text{rob}}}{du} = \frac{\theta C_1(\theta - u^2)}{(u^2 + M_{\text{adv}}C_2 u + \theta)^2},$$

positive for $u < \sqrt{\theta}$ and negative for $u > \sqrt{\theta}$, giving a unique maximum at $u = \sqrt{\theta}$, i.e. $R = \theta$. The two-root structure and ultimate boundedness then follow by reduction to Theorem 9.4 of CEK-v4 with $\Psi \to \Psi_{\text{rob}}$. $\square$

8.5 Fisher amplification — local and global

Theorem 8.6 (Fisher net-descent condition)**.** *Under $G = I_S$ with $\mu_I \le \lambda(I_S) \le M_I$:*

(i) For $\|S\| \ge s_0 > 0$, net descent under the worst-case adversary holds iff

$$E \ge \frac{M_G\big(M_{\text{adv}}\sqrt{M_I}J_{\max} + \|F_{\text{nom}}\|/s_0\big)^2}{4\sigma^2\mu_I^2}.$$

(ii) Near the target $S = 0$, linearise $S(X) = J_0(X - X^\star) + O(\|X - X^\star\|^2)$ with $J_0 = J(X^\star)$. The local condition becomes

$$\|X - X^\star\|^2 \ge \frac{M_G\big(M_{\text{adv}}\sqrt{M_I}\,\|J_0\|_{\text{op}} + \|F_{\text{nom}}\|/\sigma_{\min}(J_0)\big)^2}{4\sigma^2\mu_I^2\sigma_{\min}(J_0)^2}.$$

Proof. (i) Combine Theorem 9.3 of CEK-v4 with Cauchy–Schwarz $\langle g_X, u^\star\rangle \le M_{\text{adv}}\|g_X\| \le 2M_{\text{adv}}\sqrt{M_I}J_{\max}\|S\|$. (ii) Local linearisation in a neighbourhood of $X^\star$, removing the $\|S\|^{-1}$ singularity. $\square$

9 Extension VII — Delay / Memory Systems

9.1 Setup

The delayed canonical ODE is

$$\dot{X}(t) = F_{\mathrm{nom}}(X(t)) - g_X(X(t-\tau)) - \gamma(X(t-\tau))\,\frac{\langle F(t-\tau),\, g_X(t-\tau)\rangle}{\|g_X(t-\tau)\|^2}\, g_X(t-\tau)$$

with $\tau > 0$ a constant measurement delay. Initial history $X|_{[-\tau,0]} \in C^0$.

9.2 Lyapunov–Krasovskii descent

Definition 9.1 (LK functional)**.**

$$V(X_t) = E(X(t)) + \mu \int_{t-\tau}^{t} \|g_X(X(s))\|^2 \, ds.$$

Theorem 9.2 (LK descent inequality)**.** *Under (A1)–(A4) and the angle condition on $[t-\tau, t]$,*

$$\dot{V}(X_t) \le -(1-\gamma_\tau)\,\|g_X(t-\tau)\|^2 + \mu\,\|g_X(t)\|^2 - \mu\,\|g_X(t-\tau)\|^2 + \langle g_X(t),\, F_{\mathrm{nom}}(t)\rangle$$

where $\gamma_\tau = \gamma(X(t-\tau))$.

Proof. Standard LK differentiation; the chain-rule term $\dot{E}$ is expanded via the delayed ODE, and the cross term $\langle g_X(t),\, g_X(t-\tau)\rangle$ is absorbed into the integral term by Cauchy–Schwarz. Routine. □

Theorem 9.3 (Sufficient delay margin)**.** *The delayed canonical ODE is stable for delays*

$$\tau < \tau^\star = \frac{(1-\gamma_{\max})\rho}{L_g^2}$$

where L_g is the Lipschitz constant of $X \mapsto g_X$ on the operating set and ρ is the delay-free rate. Sharpness is not claimed.

Proof. Choose $\mu = (1-\gamma_{\max})/2$ in the LK functional. Bound $\|g_X(t) - g_X(t-\tau)\| \le L_g \tau \left\|\dot{X}\right\|_\infty$ to control the cross term, then close via Grönwall. Sufficiency only. □

9.3 Predictor recovery

Theorem 9.4 (Cheap Smith predictor: $O(\tau)$ recovery)**.** *For the Smith predictor $\hat{X}(t) = X(t-\tau) + \int_{t-\tau}^{t} F_{\mathrm{nom}}(X(s))\, ds$,*

$$\left\|X(t) - \hat{X}(t)\right\| \le \tau \cdot \|g_X + \gamma\alpha g_X\|_\infty + O(\tau^2).$$

The predictor-corrected ODE achieves

$$\rho_{\mathrm{pred}} = \rho - O(\tau).$$

Proof. The cheap predictor uses only F_{nom} over the delay window and therefore omits the correction terms of the canonical drift; the error is first order in τ. Propagating through the energy derivative gives $O(\tau)$ rate degradation. □

Theorem 9.5 (Full-drift predictor: $O(\tau^2)$ recovery)**.** *For the full-drift predictor*

$$\hat{X}_{\text{full}}(t) = X(t-\tau) + \int_{t-\tau}^{t} \Big[F_{\text{nom}}(X(s)) - g_X(X(s)) - \gamma(X(s))\alpha(X(s))g_X(X(s))\Big]\, ds,$$

$$\Big\| X(t) - \hat{X}_{\text{full}}(t) \Big\| = O(\tau^2), \qquad \rho_{\text{pred,full}} = \rho - O(\tau^2).$$

Proof. $\hat{X}_{\text{full}}$ is the first-order Taylor expansion of $X(t)$ around $X(t-\tau)$ using the full drift. The remainder is second order in τ, with Lipschitz constants L_F, L_g, L_γ entering the constant. □

Remark 9.6 (Realisability)**.** $\hat{X}_{\text{full}}$ requires evaluating $g_X(X(s))$ and $\gamma(X(s))$ across the delay window, both of which depend on the unknown true trajectory. It is therefore an idealised predictor; practical implementations require either an auxiliary state observer or a recursive approximation scheme.

9.4 Application: drone GPS dropout

F_{nom} = inertial dead-reckoning over the dropout window. Cheap predictor: $O(\tau)$ rate degradation, easily implemented. Full predictor: $O(\tau^2)$ recovery, requires an inertial-rate observer integrating the canonical drift.

10 Master Unified View

Ext.	**Domain**	**Object changed**	**Status**
CEK-v4	general	frozen kernel	locked
I Symplectic	grid, water	$g_X \to g^\star = P_{\mathcal{H}}\nabla E$	proved
I Poisson	grid	+ Casimir-tangent projection	proved
II Lie group	drone, robot	$J \to \hat{J}^*$, exp integrator	proved (local)
III Pareto	finance, robot	$g_X \to \bar{g}_X$, $\gamma \to \bar{\gamma}$	proved on support
IV Graph/fractal	grid, water	$S \to S_{\text{graph}}$, $G \to G_{\text{graph}}$, heat kernel	proved
V Kurtosis prec.	finance	$G \to G^{(\beta)}$	proved (isotropic)
VI Robust	drone	$\gamma \to \gamma_{\text{rob}}$	proved on $\mathcal{S}_{\text{rob}}$
VII Delay	drone	LK functional, predictor	proved (sufficient)

10.1 Theorem inventory

Tag	Category	Scope
CGS.1	projected descent identity	global, no rate
SP.1	conservation	exact $\dot{\mathcal{H}} = 0$
SP.2	descent identity	with explicit angle condition
SP.3	exponential rate	local, under projected coercivity
SP.4	Casimir conservation	exact $\dot{C}_i = 0$
LG.1	manifold invariance	$X(t) \in G$ for all $t \geq 0$
LG.2	descent on G	local
LG.3	exponential rate on G	local, $O(K_0 r^2)$ correction
PO.1	descent on support of $\lambda^\star$	local
PO.2	Pareto stationary convergence	global on joint sublevel
PO.3	Pareto sliding	local, requires tangency
DFL.1	spectral coercivity	global on $\mathbf{1}^\perp$
DFL.2	exponential convergence	on invariant sublevel inside $\mathbf{1}^\perp$
DFL.3	total energy descent	local; per-scale requires orthogonality
AC.1	validity range	explicit interval in β
AC.2	descent	local
AC.3	strict rate monotonicity	isotropic kurtosis, frozen sublevel
AR.1	minimax value, no interior saddle	pointwise
AR.2	implementable robust descent	on $\mathcal{S}_{\mathrm{rob}}$
AR.3	robust ultimate boundedness	local in energy
AR.4	Fisher net-descent condition	global ($\|S\| \geq s_0$) + local
MD.1	LK descent inequality	global on history
MD.2	sufficient delay margin	local; sufficiency only
MD.3	cheap predictor	$O(\tau)$ rate degradation
MD.3′	full predictor	$O(\tau^2)$ rate, idealised

10.2 Architectural commitments

1. CEK-v4 remains the locked Euclidean special case; nothing in this manuscript redefines, weakens, or replaces its frozen ODE, gradient formula, Euclidean projection, or control gain.

2. CGS-v1 is the universal architecture. Every extension is an instantiation of the CGS-v1 flow via a choice of $(D_X, \nabla\mathcal{E}_\lambda, F_{\mathrm{base}})$ or via upstream modification of $(S, G, F_{\mathrm{base}})$.

3. The projection acts on the correction field, not on the chain-rule gradient.

4. Every theorem in this manuscript carries explicit hypotheses, an explicit coercivity constant where applicable, and an explicit scope (global, local, pointwise, or restricted to an invariant subspace).

5. No heuristic survives in the theorem list. Every claim of asymptotic order, monotonicity, descent, or boundedness is proved under the stated assumptions or removed entirely.

11 Conclusion

The manuscript establishes that the Canonical Euclidean Kernel of v4 admits seven extensions to geometries beyond Euclidean space, each preserving the locked canonical structure or instantiating the broader CGS-v1 universal flow when the locked Euclidean projection is genuinely incompatible with the target geometry (as in the symplectic and Lie-group cases). The framework now operates under the discipline of narrow hypotheses with explicit constants rather than broad universal claims, and the theorems delivered match what is actually proved. The strongest sections — Hamiltonian-tangent projection (Ext. I), Lie-group canonical damping (Ext. II), graph-wavelet canonical flow (Ext. IV), and Lyapunov–Krasovskii delay analysis (Ext. VII) — constitute the most defensible parts of the framework; the kurtosis-corrected preconditioner (Ext. V), Pareto multi-objective system (Ext. III), and robust adversarial damping (Ext. VI) are stated under restricted hypotheses with explicit limitations on their scope.